ERD to Relational Schema Mapping

Naseej patient health record platform | 14 tables | 17 relationships

1. Mapping rules applied

ERD construct	Relational rule
Strong entity	Becomes one table; the entity identifier becomes the primary key (PK).
Attribute	Becomes one column; the data type is kept as drawn in the ERD.
Unique attribute	Column with a UNIQUE key (UK): USERS.email, AUTH_TOKENS.token_hash, RECORD_SHARES.token_hash.
One-to-many (1 : N)	The PK of the 1-side is copied into the N-side table as a foreign key (FK).
One-to-one, optional (1 : 0..1)	As 1 : N, with the FK also UNIQUE so at most one row points to the parent (PATIENT_PROFILES.user_id).
Optional participation (0..1)	The FK column accepts NULL (RECORD_SHARES.prep_id, HEALTH_ENTRIES.source_document_id, HEALTH_ENTRIES.source_fact_id).
Many-to-many (M : N)	Resolved by an associative table whose composite PK is made of the two FKs (SHARE_DOCUMENTS).
Dependent entity	The table takes the parent key as part of a composite PK (CONTENT_TRANSLATIONS, DOCUMENT_LINKS).

2. Relational schema

Notation: **TABLE** (primary key, *foreign key* → referenced table, *key that is both*, other columns). [UK] marks a unique key. Only the columns shown in the ERD are listed.

USERS (id, email [UK], password_hash, email_verified_at, password_changed_at)

AUTH_TOKENS (id, *user_id* → USERS, purpose, token_hash [UK], expires_at, used_at)

PATIENT_PROFILES (id, *user_id* → USERS [UK], ai_assistant_consent_at)

AI_INTERACTIONS (id, *patient_id* → PATIENT_PROFILES, kind, outcome)

CONTENT_TRANSLATIONS (*patient_id* → PATIENT_PROFILES, source_hash, target_lang, translated_text)

HEALTH_ENTRIES (id, *patient_id* → PATIENT_PROFILES, kind, title, source, review_status, *source_document_id* → DOCUMENTS, *source_fact_id* → DOCUMENT_FACTS, deleted_at)

SYMPTOM_REPORTS (id, *patient_id* → PATIENT_PROFILES, symptom_description, severity, onset_date, captured_via, deleted_at)

DOCUMENTS (id, *patient_id* → PATIENT_PROFILES, file_name, storage_path, doc_type, deleted_at)

DOCUMENT_FACTS (id, *document_id* → DOCUMENTS, patient_decision, value)

DOCUMENT_LINKS (*document_id* → DOCUMENTS, link_type, target_id, created_at)

VISIT_PREPS (id, *patient_id* → PATIENT_PROFILES, visit_date, status, content)

RECORD_SHARES (id, *patient_id* → PATIENT_PROFILES, *prep_id* → VISIT_PREPS, token_hash [UK], snapshot, expires_at, revoked_at)

SHARE_DOCUMENTS (*share_id* → RECORD_SHARES, *document_id* → DOCUMENTS)

SHARE_ACCESS_EVENTS (id, *share_id* → RECORD_SHARES, event, occurred_at)

3. Relationship mapping

Each relationship drawn in the ERD and how it is realised as a foreign key. Cardinality is read parent : child.

#	Parent	Child	Relationship	Cardinality	Mapping result
1	USERS	PATIENT_PROFILES	account_for	1 : 0..1	FK PATIENT_PROFILES.user_id → USERS.id, UNIQUE
2	USERS	AUTH_TOKENS	issues	1 : 0..*	FK AUTH_TOKENS.user_id → USERS.id
3	PATIENT_PROFILES	AI_INTERACTIONS	logs_metadata	1 : 0..*	FK AI_INTERACTIONS.patient_id → PATIENT_PROFILES.id
4	PATIENT_PROFILES	CONTENT_TRANSLATIONS	owns	1 : 0..*	FK part of composite PK (patient_id, source_hash, target_lang)
5	PATIENT_PROFILES	HEALTH_ENTRIES	owns	1 : 0..*	FK HEALTH_ENTRIES.patient_id → PATIENT_PROFILES.id
6	PATIENT_PROFILES	SYMPTOM_REPORTS	records	1 : 0..*	FK SYMPTOM_REPORTS.patient_id → PATIENT_PROFILES.id
7	PATIENT_PROFILES	DOCUMENTS	owns	1 : 0..*	FK DOCUMENTS.patient_id → PATIENT_PROFILES.id
8	PATIENT_PROFILES	VISIT_PREPS	prepares	1 : 0..*	FK VISIT_PREPS.patient_id → PATIENT_PROFILES.id
9	PATIENT_PROFILES	RECORD_SHARES	creates	1 : 0..*	FK RECORD_SHARES.patient_id → PATIENT_PROFILES.id
10	VISIT_PREPS	RECORD_SHARES	source_prep	0..1 : 0..*	Optional FK RECORD_SHARES.prep_id → VISIT_PREPS.id (nullable)
11	DOCUMENTS	DOCUMENT_FACTS	has_suggestions	1 : 0..*	FK DOCUMENT_FACTS.document_id → DOCUMENTS.id
12	DOCUMENTS	HEALTH_ENTRIES	source_document	0..1 : 0..*	Optional FK HEALTH_ENTRIES.source_document_id → DOCUMENTS.id (nullable)
13	DOCUMENT_FACTS	HEALTH_ENTRIES	confirmed_into	0..1 : 0..*	Optional FK HEALTH_ENTRIES.source_fact_id → DOCUMENT_FACTS.id (nullable)
14	DOCUMENTS	DOCUMENT_LINKS	linked_by	1 : 0..*	FK part of composite PK (document_id, link_type, target_id)
15	DOCUMENTS	SHARE_DOCUMENTS	included	1 : 0..*	M:N resolved; FK part of composite PK (share_id, document_id)
16	RECORD_SHARES	SHARE_DOCUMENTS	grants	1 : 0..*	M:N resolved; FK part of composite PK (share_id, document_id)
17	RECORD_SHARES	SHARE_ACCESS_EVENTS	records	1 : 0..*	FK SHARE_ACCESS_EVENTS.share_id → RECORD_SHARES.id

4. Table specifications

PK = primary key, FK = foreign key (→ referenced table), UK = unique key. Data types are those shown in the ERD.

USERS

Column	Type	Key
id	uuid	PK
email	text	UK
password_hash	text	
email_verified_at	timestamptz	
password_changed_at	timestamptz	

AUTH_TOKENS

Column	Type	Key
id	uuid	PK
user_id	uuid	FK → USERS
purpose	varchar	
token_hash	char	UK
expires_at	timestamptz	
used_at	timestamptz	

PATIENT_PROFILES

Column	Type	Key
id	uuid	PK
user_id	uuid	FK → USERS, UK
ai_assistant_consent_at	timestamptz	

AI_INTERACTIONS

Column	Type	Key
id	uuid	PK
patient_id	uuid	FK → PATIENT_PROFILES
kind	varchar	
outcome	varchar	

CONTENT_TRANSLATIONS

Column	Type	Key
patient_id	uuid	PK, FK → PATIENT_PROFILES
source_hash	char	PK
target_lang	char	PK
translated_text	text	

HEALTH_ENTRIES

Column	Type	Key
id	uuid	PK
patient_id	uuid	FK → PATIENT_PROFILES
kind	varchar	
title	varchar	
source	varchar	
review_status	varchar	
source_document_id	uuid	FK → DOCUMENTS
source_fact_id	uuid	FK → DOCUMENT_FACTS
deleted_at	timestamptz	

SYMPTOM_REPORTS

Column	Type	Key
id	uuid	PK
patient_id	uuid	FK → PATIENT_PROFILES
symptom_description	text	
severity	int	
onset_date	timestamptz	
captured_via	varchar	
deleted_at	timestamptz	

DOCUMENTS

Column	Type	Key
id	uuid	PK
patient_id	uuid	FK → PATIENT_PROFILES
file_name	varchar	
storage_path	varchar	
doc_type	varchar	
deleted_at	timestamptz	

DOCUMENT_FACTS

Column	Type	Key
id	uuid	PK
document_id	uuid	FK → DOCUMENTS
patient_decision	varchar	
value	varchar	

DOCUMENT_LINKS

Column	Type	Key
document_id	uuid	PK, FK → DOCUMENTS
link_type	varchar	PK
target_id	uuid	PK
created_at	timestamptz	

VISIT_PREPS

Column	Type	Key
id	uuid	PK
patient_id	uuid	FK → PATIENT_PROFILES
visit_date	date	
status	varchar	
content	jsonb	

RECORD_SHARES

Column	Type	Key
id	uuid	PK
patient_id	uuid	FK → PATIENT_PROFILES
prep_id	uuid	FK → VISIT_PREPS
token_hash	char	UK
snapshot	jsonb	
expires_at	timestamptz	
revoked_at	timestamptz	

SHARE_DOCUMENTS

Column	Type	Key
share_id	uuid	PK, FK → RECORD_SHARES
document_id	uuid	PK, FK → DOCUMENTS

SHARE_ACCESS_EVENTS

Column	Type	Key
id	uuid	PK
share_id	uuid	FK → RECORD_SHARES
event	varchar	
occurred_at	timestamptz	

Scope note. The ERD shows the active patient-owned model with its key columns only. The full schema (migrations 001 to 007, including the legacy multi-centre tables that remain unused) is described in docs/reference/database-erd.md.